OB/GYN Clinical Pathology & Metabolic Reference Master Guide

Educational Reference Data: Combined Medical Chemistry, Hematology, Urinalysis, Cytology, and Fetal Screening Guidelines.

Diagnostic Category	Test Name	Standard Adult Reference Range	Normal Pregnancy Target Range	Primary Clinical Purpose
Electrolytes & Acid-Base	Sodium (Na■)	135 – 145 mEq/L	135 – 145 mEq/L	Governs extracellular fluid levels and nerve signaling.
Electrolytes & Acid-Base	Potassium (K■)	3.5 – 5.1 mEq/L	3.5 – 5.1 mEq/L	Essential for skeletal and cardiac muscle contraction.
Electrolytes & Acid-Base	Chloride (Cl■)	98 – 107 mEq/L	98 – 107 mEq/L	Works with sodium to preserve blood pressure and fluid distribution.
Electrolytes & Acid-Base	Carbon Dioxide (CO■ / HCO■■)	22 – 29 mEq/L	18 – 26 mEq/L	Acts as a chemical buffer to maintain strict blood pH.
Renal (Kidney) Function	Blood Urea Nitrogen (BUN)	7 – 20 mg/dL	5 – 12 mg/dL	Measures waste products from routine protein breakdown.
Renal (Kidney) Function	Creatinine	0.50 – 1.10 mg/dL	0.40 – 0.80 mg/dL	Evaluates muscle tissue breakdown and general kidney clearance.
Renal (Kidney) Function	BUN/Creatinine Ratio	10:1 – 20:1	10:1 – 20:1	Differentiates between physical tissue damage and dehydration.
Renal (Kidney) Function	Uric Acid	2.5 – 6.0 mg/dL	Drops early, then rises	Evaluates purine breakdown; spikes signal preeclampsia.
Hepatic (Liver) Function	Alanine Aminotransferase (ALT)	4 – 36 U/L	4 – 36 U/L	Liver enzyme that rises rapidly during cellular tissue injury.
Hepatic (Liver) Function	Aspartate Aminotransferase (AST)	8 – 33 U/L	8 – 33 U/L	Tracks active cell destruction in liver, heart, and muscles.
Hepatic (Liver) Function	Alkaline Phosphatase (ALP)	30 – 120 U/L	Can double or triple	Identifies biliary blockage, bone disease, or placental growth.
Hepatic (Liver) Function	Gamma-Glutamyl Transferase (GGT)	5 – 40 U/L	5 – 40 U/L	Sensitive marker for bile duct irritation or toxicity.
Hepatic (Liver) Function	Total Bilirubin	0.2 – 1.2 mg/dL	0.2 – 1.2 mg/dL	Evaluates waste pigment from recycled red blood cells.
Hepatic (Liver) Function	Total Protein	6.4 – 8.3 g/dL	6.0 – 7.0 g/dL	Measures the overall sum of protein floating in serum.
Hepatic (Liver) Function	Albumin	3.5 – 5.0 g/dL	3.0 – 4.2 g/dL	Holds fluid inside blood vessels to prevent tissue swelling.
Hepatic (Liver) Function	Globulin	2.3 – 3.4 g/dL	2.3 – 3.4 g/dL	Measures circulating immune system proteins (antibodies).
Pancreatic Exocrine	Amylase	30 – 110 U/L	30 – 110 U/L	Enzyme tasked with digesting complex carbohydrates.
Pancreatic Exocrine	Lipase	0 – 160 U/L	0 – 160 U/L	Specific enzyme tracking localized pancreas inflammation.
Metabolic & Minerals	Fasting Glucose	70 – 99 mg/dL	< 95 mg/dL	Measures baseline sugar energy levels in the bloodstream.

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Metabolic & Minerals	Calcium	8.6 – 10.2 mg/dL	Drops slightly	Crucial for cell signals, bone health, and blood clotting.
Metabolic & Minerals	Phosphorus	2.5 – 4.5 mg/dL	2.5 – 4.5 mg/dL	Partners with calcium to regulate cellular energy storage.
Metabolic & Minerals	Magnesium	1.5 – 2.5 mg/dL	1.5 – 2.5 mg/dL	Controls electrical enzyme signaling and muscle movements.
Cardiovascular & Lipids	Total Cholesterol	< 200 mg/dL	Rises to 250–300+	Building blocks for cellular membranes and vital hormones.
Cardiovascular & Lipids	HDL ("Good") Cholesterol	> 50 mg/dL	> 50 mg/dL	Transports cholesterol away from arteries back to the liver.
Cardiovascular & Lipids	LDL ("Bad") Cholesterol	< 100 mg/dL	Variable	Tracks circulating particles linked to arterial plaque buildup.
Cardiovascular & Lipids	Triglycerides	< 150 mg/dL	Can double or triple	Primary fat storage particle used for systemic energy.
Cardiovascular & Lipids	hs-CRP (Inflammation)	< 1.0 mg/L	< 1.0 mg/L	Measures systemic inflammation to predict cardiac vessel risk.
Cardiovascular & Lipids	High-Sensitivity Troponin	< 14 ng/L	< 14 ng/L	Premier emergency marker to rule out acute heart damage.
Hematology (CBC)	White Blood Cells (WBC)	4.5 – 11.0 x10³/µL	5.6 – 14.9 x10³/µL	Monitors the body's acute immune responses and screens for infections.
Hematology (CBC)	Red Blood Cells (RBC)	3.80 – 5.10 x10■/µL	3.20 – 4.50 x10■/µL	Evaluates the concentration of oxygen-carrying red blood cells.
Hematology (CBC)	Hemoglobin	11.7 – 15.5 g/dL	11.0 – 14.0 g/dL	Directly measures oxygen-binding proteins to screen for maternal anemia.
Hematology (CBC)	Hematocrit	35.0% – 45.0%	28.0% – 40.0%	Represents the packed cell volume percentage of red blood cells.
Hematology (CBC)	Platelets	140 – 400 x10³/µL	100 – 330 x10³/µL	Assesses natural blood-clotting capability to prevent hemorrhaging.
Blood TYPE Compatibility	ABO Group	Type A, B, AB, or O	Type A, B, AB, or O	Identifies the basic biological blood type group of the individual.
Blood TYPE Compatibility	Rh Factor Type	Positive or Negative	Rh-Positive	If Rh-Negative, identifies the risk for maternal-fetal blood group conflict.
Blood TYPE Compatibility	RBC Antibody Screen	Negative	Negative	Checks for irregular immune antibodies that could attack fetal red blood cells.
Infectious Serology	Rubella Antibody (IgG)	Negative (Non-Immune)	Positive (Immune)	Confirms existing vaccine-induced or natural protective immunity to German Measles.
Infectious Serology	Hep B Surface Ag (HBsAg)	Negative (Non-Reactive)	Negative (Non-Reactive)	Screens for active maternal Hepatitis B infection to protect newborns at delivery.
Infectious Serology	Syphilis Serology (RPR)	Negative (Non-Reactive)	Negative (Non-Reactive)	Screens for active syphilis infection to prevent severe congenital defects.

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Infectious Serology	HIV-1 / HIV-2 Screen	Negative (Non-Reactive)	Negative (Non-Reactive)	Identifies HIV status to allow early antiviral therapy to prevent vertical transmission.
Urinalysis (UA)	Urine Protein	Negative to Trace	1+ to 4+ Protein	Key screening marker for preeclampsia or severe renal stress.
Urinalysis (UA)	Urine Glucose	Negative	Positive (Glucosuria)	Detects renal threshold spillover, indicating gestational diabetes.
Urinalysis (UA)	Leukocyte Esterase & Nitrites	Negative	Positive	Rapid screen for asymptomatic or acute Urinary Tract Infections (UTIs).
Urinalysis (UA)	Urine Culture & Sensitivity	No Growth	>100,000 CFU/mL	Identifies the exact bacterial strain to guide safe antibiotic selection.
Urinalysis (UA)	Urine hCG	Negative (Unless pregnant)	Positive	Qualitatively detects human chorionic gonadotropin to verify pregnancy.
Cytology & Swabs	Liquid-Based Pap Smear	Negative for Malignancy	ASCUS, LSIL, HSIL	Screens cervical cells for precancerous changes or carcinoma.
Cytology & Swabs	High-Risk HPV Co-test	Negative	Positive	Detects high-risk oncogenic viral strains (e.g., HPV 16 and 18).
Cytology & Swabs	Vaginal Wet Mount (Prep)	Normal Lactobacilli	Clue cells, hyphae, trichomonads	Diagnoses Bacterial Vaginosis (BV), yeast infections, or parasites.
Cytology & Swabs	Chlamydia & Gonorrhea NAAT	Negative	Positive	Molecular screening for common, often asymptomatic STIs.
Cytology & Swabs	Group B Strep (GBS) Culture	Negative (at 36-37 wks)	Positive	Dictates the need for IV antibiotics during labor to protect the newborn.
Fetal Screens	Cell-Free DNA (NIPT)	Low Risk	High Risk	Analyzes fetal DNA in maternal blood for Trisomies (13, 18, 21).
Fetal Screens	Fetal Fibronectin (fFN)	Negative	Positive	Evaluates the immediate, short-term risk of preterm delivery.
Fetal Screens	Nitrazine & Ferning Test	Negative (pH 4.5–6.0)	Positive (pH > 7.0 / Fern)	Confirms Premature Rupture of Membranes (PROM) (broken water).

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.